

## **Teaching Statement**

I teach students as emerging engineering professionals. I aim to develop the qualities I value in a colleague: a commitment to safety, sound engineering judgment, careful use of evidence, and clear communication. I structure learning experiences so that students move from guided practice toward independent inquiry while considering technical performance, cost, scalability, reliability, and ethical implications.

## **Learning to Ask Good Questions**

I encourage students to approach unfamiliar problems with curiosity and to use evidence rather than assumptions to guide their decisions. In RES 100, students begin with a topic that matters to them and develop it into a focused, evidence-based research question. They learn to search academic databases strategically, evaluate sources, identify relevant researchers, and define a manageable scope.

When one group struggled to narrow its topic, I guided them through a targeted literature search rather than suggesting a topic for them. This helped them understand the existing work, identify a meaningful scope, and develop confidence in pursuing the question independently. Their work culminates in a team mock-poster presentation that brings together research, collaboration, and scientific communication.

I use assessment as scaffolding rather than only as a measure of final performance. Quizzes and short assignments develop foundational skills such as database searching, research ethics, and the benefits, risks, and ethical considerations of AI in research. Each activity contributes to the final poster, allowing students to receive feedback while there is still time to revise.

## **Learning to Make Engineering Decisions**

In project-based teaching, I encourage students to move beyond asking whether something can be built toward asking whether it is an appropriate solution to the problem.

As a prototyping mentor in EE311, I help student teams turn an initial idea into a testable engineering proposal. I use structured questions to prompt students to identify the intended user and decision, define what the system should produce, consider where a simple implementation may be misleading, and select a meaningful test or comparison. Teams then share their reasoning and receive feedback as they narrow the scope of their prototype and plan a convincing final demonstration.

This helps students see that engineering decisions should be grounded in evidence and constraints, not novelty alone. Students learn to consider not only whether a system works, but also how its performance should be evaluated and whether the resulting evidence supports a conclusion.

## **Building, Testing, and Troubleshooting**

Laboratory teaching is an important setting for developing independent technical reasoning. In KIE1001 laboratory teaching, I supported students as they progressed from basic electrical components and breadboarding to combinational and sequential logic, circuit and electronics experiments, and prototype development.

I ask students to prepare by reading schematics, then emphasise correct assembly, safe practice, and functional verification. When a circuit does not work as expected, I guide students through systematic troubleshooting before providing an answer. We consider possible causes, identify what evidence would distinguish between them, and test the most plausible explanation.

In this way, an unexpected result becomes an opportunity to develop diagnostic reasoning and confidence rather than simply an error to correct. Students learn that a failed experiment can provide valuable information when approached systematically.

### **From Guided Work to Independent Research**

My research mentoring follows the same progression from structured guidance toward independent judgment. I support students in experimental planning, safe laboratory practice, equipment operation, troubleshooting, data analysis, and scientific communication, while gradually transferring responsibility for decisions to the student.

This has included mentoring a Master's researcher through the development of fiber Bragg grating sensors and optical interrogation systems, from optical-component design and laboratory execution to data analysis and paper preparation. She subsequently progressed to PhD study at KFUPM.

I have also mentored undergraduate research projects involving LiDAR-based systems, specklegram fiber sensors, and fiber Bragg grating sensing for underwater temperature monitoring. These projects have involved experimental design, data analysis, poster presentations, and manuscript or conference-submission preparation.

### **Helping Students Find Their Path**

Students enter engineering with different levels of prior experience in mathematics, programming, laboratory work, and research. I therefore use multiple entry points into technical concepts, combining conceptual explanation, practical demonstration, guided questions, experimentation, and data analysis.

In my RES 100 section for female engineering students, I also emphasise that research and engineering can lead to varied careers in academia, industry, and policy. I extend this encouragement through virtual outreach sessions with female high-school students in Saudi Arabia, where I discuss research opportunities and pathways into engineering.

I want students to see themselves as capable participants in engineering and to understand how the skills they are developing can open different pathways for them.

## Looking Ahead

I can contribute to undergraduate teaching across electrical and electronic engineering, including circuits, electronics, digital systems, signals and systems, digital signal processing, control, electromagnetics, communications, electrical measurements and instrumentation, and associated laboratory courses. At advanced undergraduate and postgraduate level, my teaching interests include optical fiber communications, optics and photonics, optoelectronic devices, fiber-optic sensing, industrial instrumentation, and research methodology.

I would like to develop research-integrated laboratory and project experiences in which students work with authentic engineering problems, experimental measurements, and real datasets. In particular, I am interested in connecting physical experimentation with computational analysis and machine-learning methods, allowing students to experience the iterative process of measurement, troubleshooting, validation, and scientific communication that characterises engineering practice.

Ultimately, I want students to see engineering as both a body of knowledge and a way of making responsible decisions under real-world constraints. Through teaching and mentoring, I aim to help them develop the judgment, practical skills, and confidence to take a problem from an initial idea to a defensible solution.